

FE BAR SUBMISSION · BUSINESS VALUE BRIEF

Hire Right

AI-powered predictive hiring for Jackson & Jackson HR Digital

From raw applicant data to a governed, conversational hiring decision — one connected journey on Databricks.

THE PROBLEM

Hiring senior HR leaders is slow, subjective, and expensive to get wrong.

- ◆ **45+ days** to fill a director-level role — top candidates are gone before the shortlist is set.
- ◆ Decisions rest on **gut feel and inconsistent scorecards**, creating bias and compliance exposure in a regulated industry.
- ◆ A single **mis-hire at the leadership level costs 1.5–3× salary** in severance, lost productivity, and re-recruiting.
- ◆ Applicant data, resumes, and analytics live in **disconnected tools** — no single, governed source of truth.

THE OUTCOME — LEAD WITH IMPACT

Faster, fairer, defensible hiring decisions.

60%

Faster time-to-fill

Ranked shortlists on day one, not week six

~30%

Fewer mis-hires

8-factor ML scoring lifts quality of hire

15 hrs

Saved per requisition

Automated screening & resume Q&A

100%

Auditable & consistent

Governed scoring, masked PII, tracked notes

Projected impact for the J&J HR Digital prototype, expressed in the KPIs the CHRO and TA leadership already report on.

WHAT THE HR TEAM GETS

A decision cockpit for every requisition.

Ranked candidates

Every applicant scored across 8 weighted dimensions with a clear total & stage.

Data-science recommendation

One click runs the ML prediction and returns a hire recommendation with rationale.

Resume Q&A (RAG)

Ask about any candidate's experience — grounded in their actual resume.

Genie analytics

"Who are the top 3 for the CPO role?" answered instantly over governed data.

HR notes

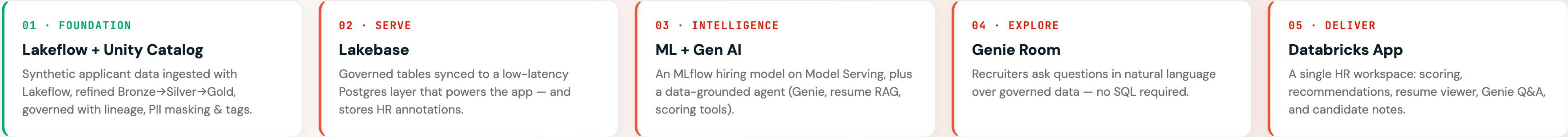
Add durable, audit-trailed annotations to a candidate record — saved to Lakebase.

Snappy & always-on

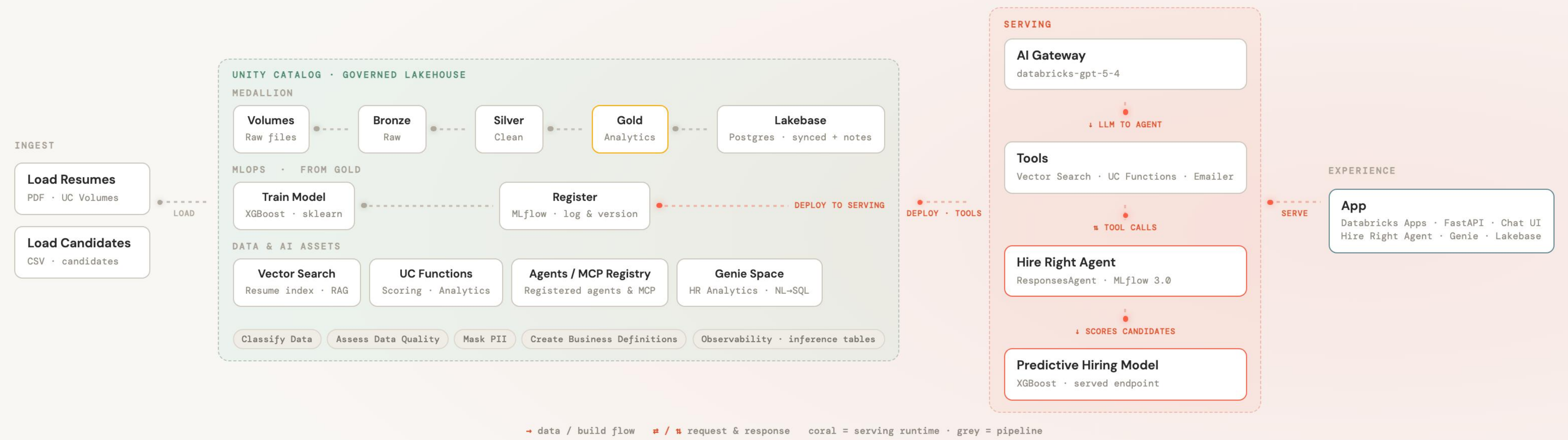
Low-latency reads from Lakebase keep the whole workspace instant.

HOW IT WORKS

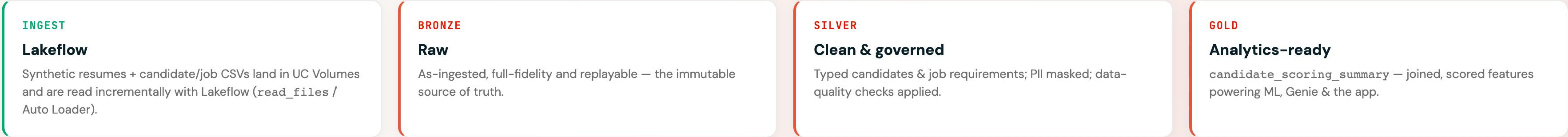
One connected journey — data to decision on Databricks.



End-to-end on Databricks — data to decision.



A governed medallion, ingested with Lakeflow.



Built for a regulated industry from day one.

PII protected

Column masking & sensitivity tags on candidate data enforced in Unity Catalog.

Full lineage

Every field traceable Bronze→Gold→Lakebase for audit and explainability.

Evaluated AI

The agent is scored with MLflow evaluation and an LLM-judge before it ships.

Accountable notes

Every HR annotation is attributed and time-stamped in a transactional store.

The same governance that protects candidates also gives the CHRO a defensible, explainable hiring process.

TECHNICAL • OPERATIONAL SERVING

Lakebase turns the lakehouse into an app backend.

Synced tables

Gold candidate & job tables synced Delta→Postgres for millisecond, app-facing reads.

Transactional notes

A native `candidate_annotations` table captures HR notes, joined to candidate data.

Powers the app

Candidate cards, job views and notes all served from Lakebase — a snappy UX.

Secure by default

The app connects with short-lived OAuth credentials, governed in Unity Catalog.

From labeled history to a served hire model.

01 · TRAIN

Three models

LogReg, Random Forest & Gradient Boosting on the 8 scoring features (Silver).

02 · TRACK

MLflow

Every run logged — accuracy, AUC, F1 and feature importances, fully comparable.

03 · REGISTER

@champion

The best model is promoted to @champion in Unity Catalog.

04 · SERVE

Model Serving

A scale-to-zero real-time endpoint returns the hire / no-hire decision.

A ResponsesAgent that reasons with your data.

query_genie

Natural-language analytics over the governed Gold layer.

search_resumes

Vector Search RAG grounded in candidate resumes.

predict_hiring_score

Calls the served ML model for a hire recommendation.

send_email

Mailgun-backed mailer — powers the AI offer letter.

MLflow 3.0 ResponsesAgent with a Python tool-calling loop — and tool calls **stream live into the app** as they run, so HR sees the agent's work in real time.

Genie — ask the data in plain English.

- ◆ **NL→SQL** over the governed Gold layer, guided by business semantics & metric definitions.
- ◆ Powers both the app's **Genie panel** and the agent's **query_genie** tool.
- ◆ No SQL required — recruiters get instant answers that stay **governed and auditable**.

Quality gates before the agent ships.

MLflow evaluation

Built-in and custom scorers run against a curated eval set.

LLM-as-judge

An LLM judge scores tone & consistency on every response.

Tracing

Each tool call is traced for debugging and production observability.

Gated deploy

Only agents that clear the quality bar are promoted to serving.

WHY DATABRICKS

Data, AI, and the app — on one governed platform.

- ◆ **No integration tax:** ingestion, governance, ML, Gen AI, analytics, and the app all live together.
- ◆ **Lakebase** turns governed lakehouse tables into a low-latency application backend — no separate database to sync and secure.
- ◆ **Unity Catalog** governs data *and* AI, so the whole solution is auditable end-to-end.
- ◆ **Faster to value:** this entire prototype was built as one connected journey, not stitched from point tools.

THE ASK

Let's hire right, together.

A working, governed, AI-powered hiring solution — ready to pilot on a live requisition and prove the numbers on your data.

- ◆ Pilot on one open role and baseline time-to-fill & quality of hire
- ◆ Extend the model and Genie room to your historical hiring data
- ◆ Roll the app out to the TA team behind Unity Catalog governance